Symphonix Patient Identity Closed Loop

Architecture decision, requirements, use cases, traceability matrix, and test harness

Field	Value
Canonical identifier	SPID - Symphonix Patient Identifier
Identifier system	https://fhir.symphonix.health/Id/patient
NHS Number role	External alias only, never the internal master patient id
Requirements	18
Scenarios	12

Executive Decision

Symphonix Health should standardise BulletTrain, Shared HealthRecord, Client Registry, and all patient-data siblings on SPID. SPID has the 10-digit modulus-11 shape used by the UK NHS number format, but it is a Symphonix identifier with a Symphonix FHIR system. The Client Registry is the only issuing authority. Other services maintain local projections and fail closed when patient identity is invalid, missing, under review, restricted, or deceased.

Architecture Summary

- Client Registry owns SPID allocation, aliases, lifecycle status, duplicate/confused-record review, merge/link, restricted status, and death status.
- Shared HealthRecord and siblings own clinical/domain records but key patient context by SPID.
- Patient creation is propagated through a transactional outbox and patient.identity.* events.
- Agents can detect, explain, and recommend identity corrections, but deterministic services and human approval own safety-critical state changes.
- Closed-loop reconciliation compares registry and subscriber projections and replays events or opens review tasks for drift.

NHS/PDS Lessons Applied

NHS Pattern	Symphonix Design Rule
PDS master demographics	Use Client Registry as the Symphonix master patient identity service.
Duplicate records	Merge confirmed duplicates to one SPID and retain aliases.
Confused records	Put records under review and require NBO-style human approval before correction.
Sensitive flag / stop-noting	Represent as restricted identity status and suppress location/contact disclosure.
Death status	Stop active care loops and notifications without deleting the record.
Incorrect death status	Correct through audited reinstatement or death-status correction events.

Core Requirements

ID	Priority	Family	Requirement	Scenario Count
SPID-FR-001	P0	Master Patient Identity	Client Registry is issuing authority	2
SPID-FR-002	P0	Master Patient Identity	NHS-style checksum shape	2
SPID-FR-003	P0	Master Patient Identity	Not an NHS number	2
SPID-FR-004	P0	Fhir Canonical	FHIR Patient identity binding	2
SPID-FR-005	P1	Fhir Canonical	Canonical envelope carries patient identity	1
SPID-FR-006	P0	Propagation	Creation event propagation	2
SPID-FR-007	P0	Propagation	Subscriber projection contract	2
SPID-FR-008	P0	Safety Controls	Reject unknown or inactive patient writes	3
SPID-FR-009	P0	Propagation	Idempotent outbox delivery	2
SPID-FR-010	P0	Confused Records	Confused record workflow	1
SPID-FR-011	P0	Confused Records	Merge and link audit	2
SPID-FR-012	P0	Restricted Records	Stop-note and sensitive status	1
SPID-FR-013	P0	Deceased Patients	Death status propagation	1
SPID-FR-014	P0	Audit	Identity lifecycle audit	3
SPID-FR-015	P0	Agent Governance	Agent recommendations require deterministic guardrails	2
SPID-FR-016	P1	Reconciliation	Closed-loop reconciliation	1
SPID-NFR-001	P0	Security Privacy	Minimum disclosure	1
SPID-NFR-002	P0	Operability	Migration without patient-record loss	1

Use Case Scenarios

Use Case	Type	Priority	Subsystem	Title
patient_identity_POS_0001	Positive	P0	client-registry	Create patient with registry-issued SPID
patient_identity_POS_0002	Positive	P0	shared-health-record	Creation event upserts Shared HealthRecord projection
patient_identity_POS_0003	Positive	P1	bridge-sdk	FHIR resource carries canonical SPID
patient_identity_NEG_0004	Negative	P0	patient-data-service	Clinical write with unknown patient is rejected
patient_identity_EDG_0005	Edge	P0	client-registry	Confused record enters human approval workflow
patient_identity_EDG_0006	Edge	P0	patient-data-service	Restricted patient suppresses location and automated contact
patient_identity_EDG_0007	Edge	P0	closed-loop-care	Death status stops active care loops without deleting record
patient_identity_EDG_0008	Edge	P1	reconciliation	Closed-loop reconciliation repairs projection drift
patient_identity_NEG_0009	Negative	P0	agent-governance	Agent identity mutation without approval is blocked
patient_identity_POS_0010	Positive	P0	migration	Legacy identifiers migrate to one SPID with aliases
patient_identity_NEG_0002	Negative	P0	client-registry	Invalid SPID checksum is rejected
patient_identity_NEG_0003	Negative	P0	bridge-sdk	NHS Number namespace is rejected as internal identity

Implementation Plan

Phase	Theme	Action	Priority
1	Shared contract	Adopt bridge_sdk.patient_identity, event schemas, and envelope patient metadata.	P0
2	BulletTrain registry and SHR	Make Client Registry the SPID authority and stop local PAT-* creation in SHR.	P0
3	Sibling migration	Replace internal nhs_number and PAT-* identifiers with SPID while preserving aliases.	P0
4	Closed-loop governance	Add reconciliation, human approval, audit, and agent recommendation guardrails.	P1

Harness

Fast contract suite:

```
python -m pytest tests/test_patient_identity.py
tests/harness/patient_identity/test_patient_identity_contract.py -q
```

Post-migration static enforcement:

```
$env:SYMPHONIX_PATIENT_IDENTITY_ENFORCE='1'; python -m pytest
tests/harness/patient_identity/test_patient_identity_contract.py -q
```

Sources

Source	URL
NHS PDS	https://digital.nhs.uk/services/personal-demographics-service
PDS National Back Office	https://digital.nhs.uk/services/personal-demographics-service/national-back-office-for-the-personal-demographics-service
PDS data quality incidents	https://digital.nhs.uk/services/personal-demographics-service/national-back-office-for-the-personal-demographics-service/data-quality-incidents
Access restriction / S-flag	https://digital.nhs.uk/services/personal-demographics-service/restricting-access-to-a-patients-demographic-record
NHS number format	https://digital.nhs.uk/services/personal-demographics-service/nhs-number
NHS Data Dictionary modulus-11	https://v2.datadictionary.nhs.uk/data_dictionary/data_field_notes/n/nhs_number_de.asp%40shownav%3D0.html
NHS Notify death/flag suppression	https://webcms.prod.nhsnotify.national.nhs.uk/using-nhs-notify/recipients-you-cannot-message
Records Management Code	https://transform.england.nhs.uk/information-governance/guidance/records-management-code/records-management-code-of-practice/